Task 4-5: UR5 transfer with an upright tool

Scene design and planning method

The robot transfers between two fixed, upright joint configurations around three staggered boxes. The tighter variant shifts both side pillars inward by 40 mm and increases the right pillar height by 40 mm. Every run starts from the same colliding joint-space line, using the Task 3 PoE backend. The feasibility witness used in scene design is never supplied to the optimizer.

Task 4 optimizes obstacle and smoothness cost. Task 5 additionally penalizes the L1 difference between the tool y axis and world +z. Rotation about the upright axis remains free. The selected mode is **projected**, lambda 1000. Projection, if selected, uses the PoE direction Jacobian and damped least squares; failed projections are rejected.

Frozen parameters and acceptance

61 waypoints; 64 rollouts per round; 200 iterations per restart; 3 restarts; noise 0.55 rad; planner budget 300 s. A restart is triggered after 10 rounds without improvement. The budget includes iteration evidence writes and is checked between iterations. Development uses seeds 5112-5114 and the predeclared ordered parameter search. Formal seeds 5120-5129 were excluded from tuning.

Success requires unchanged endpoints, joint limits, conservative clearance at least 35 mm, no world or self collision, and (Task 5) maximum tilt at most 5 degrees. Independent dense checks limit every joint increment to .0125 rad. These are discrete checks. The optimizer preserves feasible candidates ahead of cheaper infeasible candidates.

Original Task 1-3 protection

The original frozen results were preserved. Fresh Windows runs passed strict replay and the new hook preserved Task 1-3 trajectories and acceptance histories bit for bit. Linux-to-Windows sample roundoff was separately bounded at 8.14e-14, with identical random states.

Formal results and failure accounting

50 completed nominal cases; two scenes; ten seeds per condition. Counts and runtime include every completed planning failure. Quality summaries include successful runs only; technical interruptions are accounted for separately.

50 completed nominal cases plus 1 archived interrupted attempt. Full-output success: 20/50 complete cases, 20/51 observed attempts. The interrupted seed 5122 ablation was repeated unchanged; no completed planning failure was removed. Its partial timing is censored, so the runtime table uses completed cases.

Scene / mode	Success	Median time (s)	Median max tilt (deg)	Median gap (mm)
main_none	10/10	9.45	108.37	48.35
main_projected	0/10	149.18	-	-
variant_none	10/10	8.58	117.24	49.85
variant_projected	0/10	140.08	-	-
main_penalty	0/10	276.21	-	-

Paired successful seeds

main: n=0 common successful seeds. No paired quality or planning-time comparison is available.

variant: n=0 common successful seeds. No paired quality or planning-time comparison is available.

Failure labels

All constrained runs, including failures: 20/20 meet the upright check; 20/20 fail self-collision checks; 0/20 pass all acceptance gates.

main_none: none.

main_projected: self_collision=10.

variant_none: none.

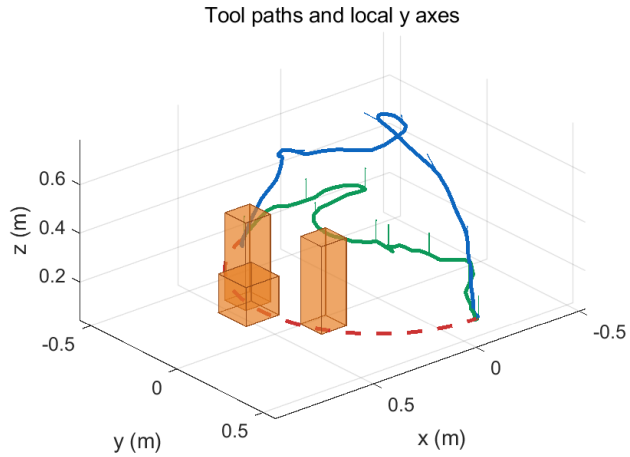
variant_projected: self_collision=10.

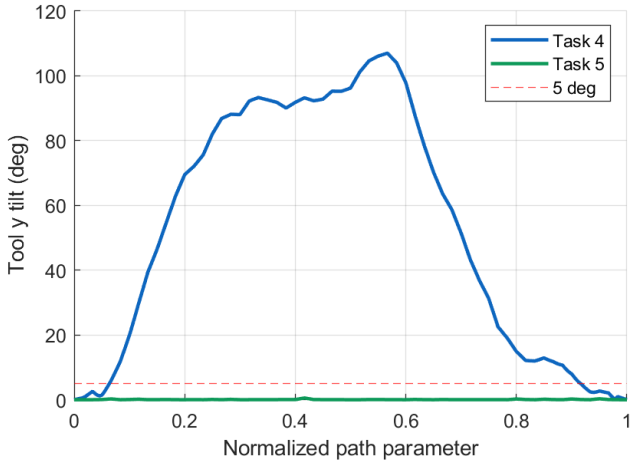
main_penalty: proxy_margin=6, orientation=10, self_collision=10.

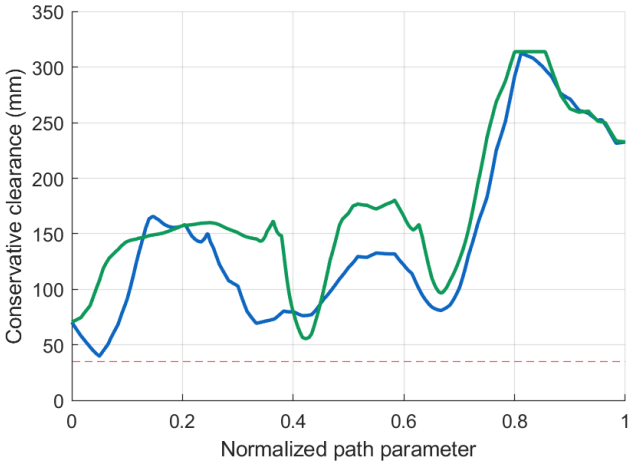
Labels can overlap. Failed runs remain saved and counted. Low cost alone is insufficient: a trajectory can violate dense clearance, self collision, or the orientation threshold. Recorded failure gates establish what failed; a specific optimizer failure mechanism requires further diagnosis.

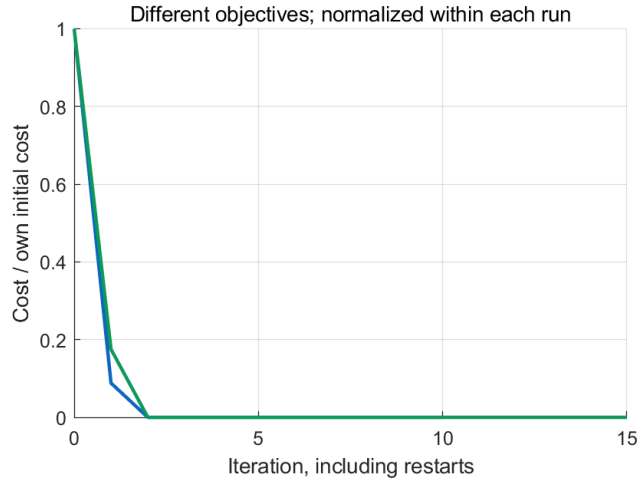
Different weighted objective totals are not compared as performance. Common-seed pairs have the same raw random conditions; projection changes realized perturbations and computation. Runtime describes this machine and software configuration.

Main scene: verified demonstration







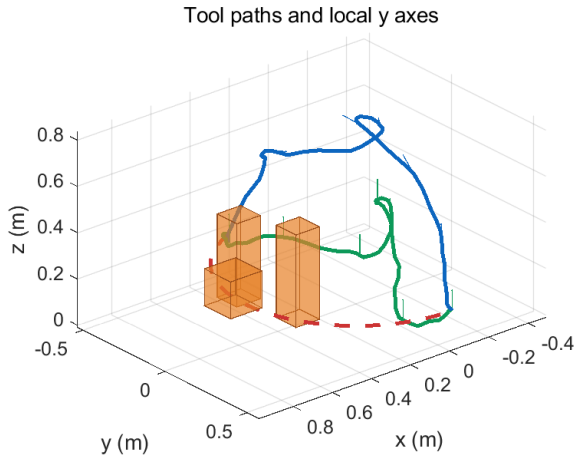


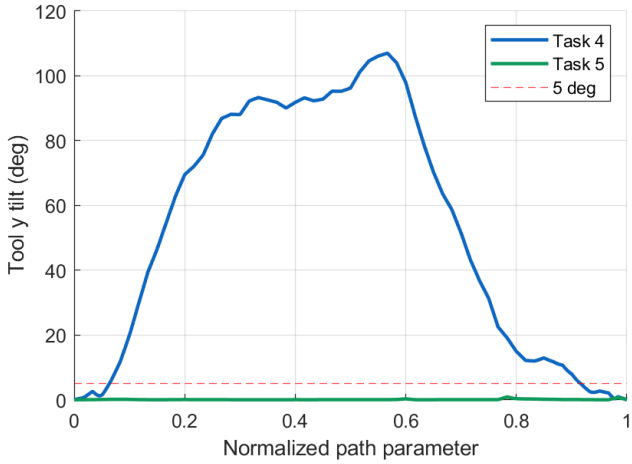
Piecewise linear path progress; arrows show the local tool y axis. The red clearance line is 35 mm, and the angular gate is 5 degrees. The cost plot normalizes each condition by its own initial cost.

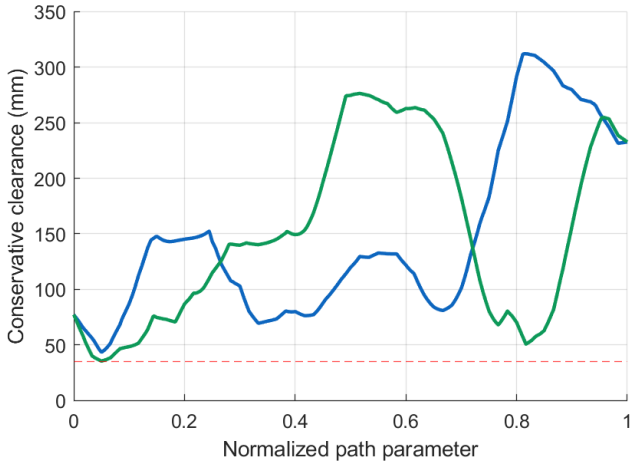
Task 4, seed 5113: gap 39.89 mm, mesh gap 126.12 mm, max tilt 106.994 degrees, 453 dense states.

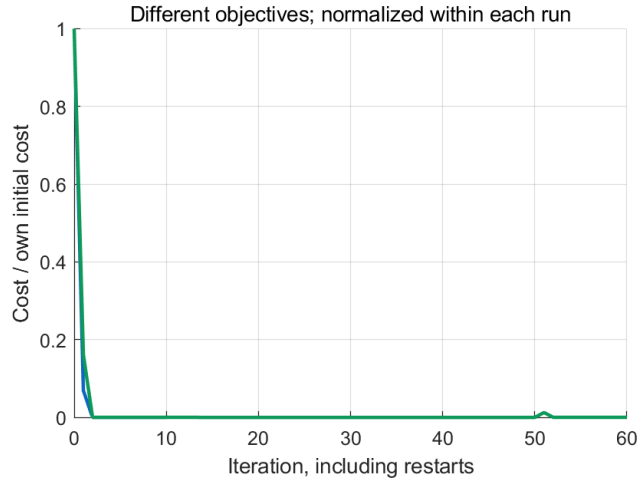
Task 5, seed 5113: gap 55.67 mm, mesh gap 123.44 mm, max tilt 0.528 degrees, 519 dense states.

Variant scene: verified demonstration









Piecewise linear path progress; arrows show the local tool y axis. The red clearance line is 35 mm, and the angular gate is 5 degrees. The cost plot normalizes each condition by its own initial cost.

Task 4, seed 5113: gap 43.33 mm, mesh gap 126.53 mm, max tilt 106.994 degrees, 453 dense states.

Task 5, seed 5113: gap 35.34 mm, mesh gap 116.67 mm, max tilt 0.868 degrees, 573 dense states.

Contribution and limitations

Member C adds scenes, upright cost/projection, unified acceptance, independent verification, paired experiments and delivery materials. The Task 1-3 PoE model, envelopes and STOMP stages are reused. No payload, dynamics, continuous collision guarantee, or physical timing is modeled.